Input to Quadratic Equation Solver

Put quadratic equation in the form:

$$\frac{1}{2}\lfloor X \quad P \quad X \rfloor^{\top} q \quad X$$

where:

X *is a Vector of size* $d+1$,

P *is a Matrix of size* $d \times d$,

qT *is a Vector of size* $1 \times d$,

?

Put all ≤ , ≥ constraints in the form:

$$G\,x \geq h$$

where:

m *is Number of constraints* ,

X *is a Vector of size* $d+1$,

G *is a Matrix of size* $m \times d$,

h *is a Vector of size* $m \times 1$,

Put all *Equal "="* constraints in the form:

$$A\,x = b$$

where:

m *is Number of constraints* ,

X *is a Vector of size* $d+1$,

A *is a Matrix of size* $m \times d$,

b *is a Vector of size* $m \times 1$,